Sentiment Analysis

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Text Processing
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Text Processing: Overview

- Introduction [RG]
- Text encoding [RG]
- Information Retrieval [RG]
- Text Compression [RG]

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- Introduction [RG]
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- Text Compression [RG]

- **Sentiment Analysis [CS]**
- **Information Extraction [CS]**
- **Introduction to Deep Learning for Text Processing [CS]**

Text Processing: SA and IE Assessment

COM3110: 10 Credit module:

- Assignment 1 [25%]
- **Assignment 2 [25%]**
- Exam [50%]

COM4115: 15 Credit module:

- Assignment 1 [25%]
- **Assignment 2 [25%]**
- Exam [50%]

Assignment 2:

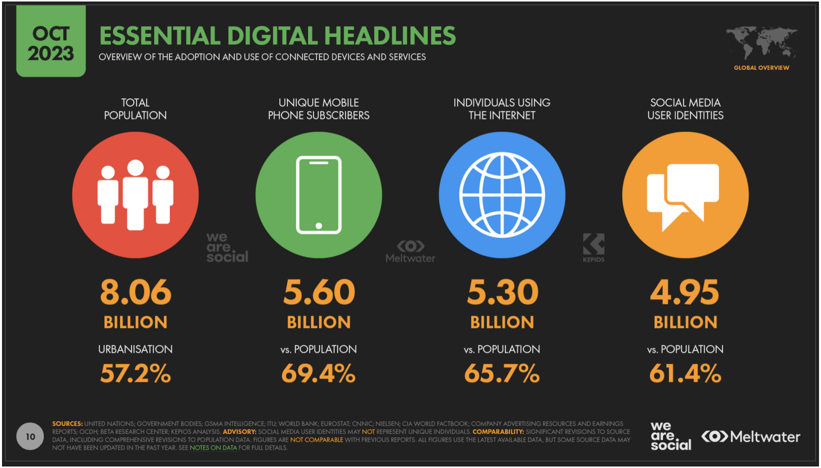
- Handout on **Friday, 24th November**
- Deadline: **Friday, 15th December @ 3pm**

Sentiment Analysis: motivations, definitions and more

Learning objectives

- Explain the SA task, including its main concepts and motivations
- Explain the differences between subjective and factual data
- Explain and apply the Bing Liu's model for SA
- Define two granularity levels for SA

The world is digital!



Source: `https://datareportal.com/global-digital-overview`

The world is digital!

People express their **emotions**, **sentiments** or **opinions**

- comments on products (Amazon, Rakuten)
- comments on movies (Rotten Tomatoes, IMDB, Youtube)
- experience in restaurants (Yelp!, Trip Advisor)
- social media (Facebook, Twitter, Instagram, LinkedIn, Reddit, Flickr)

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- **Very large quantity of information**
  - Not feasible for manual annotation
  - Companies want business intelligence!

General goal of SA

Extract **emotions**, **sentiments** or **opinions**

- Expressed by humans in **texts**
- Use that information for business or intelligence purposes

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- Customers: should you buy this washing machine?
- Prediction: who will win next election?

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Sentiment analysis ≈ opinion mining

- considered as equivalent in our case...
- ...although a sentiment does not necessarily express an opinion!

Importance of SA

To make decisions!

- **Politics:** can replace surveys, polls, etc.
- **Market:** complete/verify/correct vendor's advice
- **Television:** decide whether to continue or stop a TV show
- **Individuals:** decide whether to buy a product or not / decide whether to vote for a politician or not / etc

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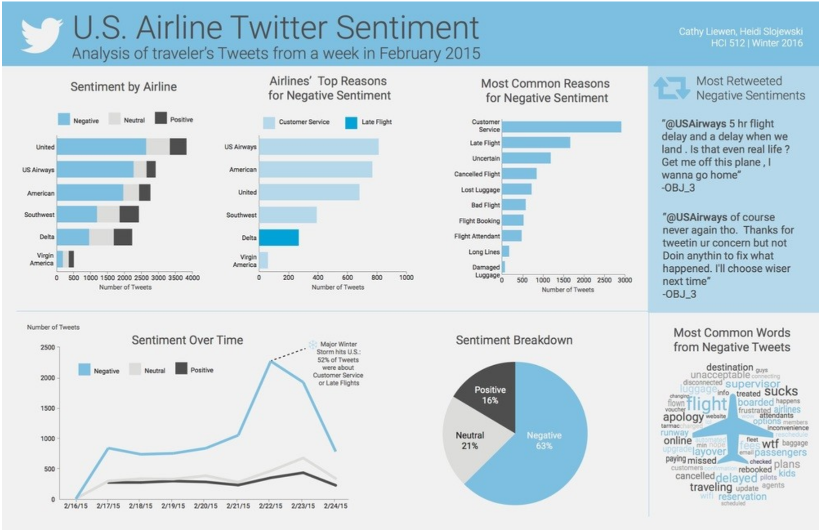
Because

- people express themselves on the web on any subject
- provides an estimate of the global opinion
- can be done at large scale

Example: social media monitoring



Source: `https://www.heidislojewski.com/blog/2016/3/11/m301ztb9orke84cuhwc37j1wwumn6n`

An example + concepts

An example

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- "@JetBlue" and "Late Flight flight experience": the **target of opinions**
- "2 hours on runway, no wifi & tv not working properly" is the **objective** part
- "I" is the **holder of opinion**

Target of opinions

A **person**, **event**, **organisation**, or **topic**

- Represented as a hierarchy of **components** having a set of **attributes**
  - let's call them both **features**
- An **opinion** is expressed about any **feature**

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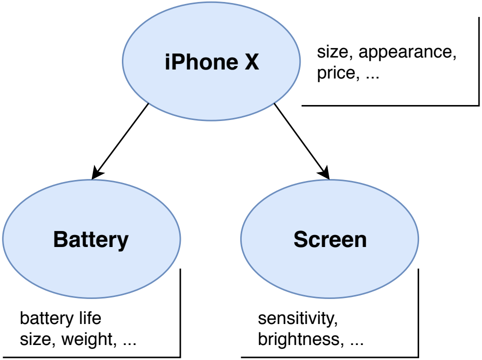

Factuality vs. Subjectivity

Factual vs. subjective data

Facts

something that is known to have happened or to exist, for which proof or information exists

- can be expressed using **keywords**
- less/not subject to interpretation
- might be difficult to identify though!

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Opinions / subjective data

a thought, belief, judgment about something or someone

- no proof required
- can be hard to express with keywords

Factual vs. subjective data: examples

What do people think of

1. *@SouthwestAir just did, thank you*

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5. *It costed 500 dollars.*

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5. *It costed 500 dollars.*
6. *It costed 500 dollars!!!*

Subjectivity analysis

Subjectivity classification is often the first step for sentiment analysis.
Aims: decide whether a text is **objective** of **subjective**

- **objective**: It costed 500 dollars.
- **subjective**: this Late Flight flight experience was the worst.

Subjectivity analysis

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- **objective**: It costed 500 dollars.
- **subjective**: this Late Flight flight experience was the worst.

However, it is not always as simple as that...

- **objective** sentences can express opinion **indirectly**:
  - *My phone broke in the second day.*
- **subjective** sentences **do not always** express positive of negative opinions:
  - *I think they will refund us.*

Bing Liu's model for SA

Bing Liu's model for Sentiment Analysis

An **opinion** is a quintuple $(o_j, f_{jk}, so_{ijkl}, h_i, t_l)$, where:

- o_j is the targeted **object** of opinion. Also called **entity**.
- f_{jk} is a **feature** of the object o_j. Also called **aspect**.
- so_{ijkl} is the **sentiment value**
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- more granular ratings ("slider")

Example

On 25th Oct. 2018 , John Doe wrote:

" I bought the new iPhone a few days ago. It was such a nice phone . The touch screen was really cool . The voice quality was clear too. Although the battery life is not long , that is ok for me. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive , and wanted me to return it to the shop."

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- f_{jk}: phone, touch screen, voice quality, battery life, price

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- time t_l : 25/10/2018

Sentiment Analysis

Opinion mining

- Discover all quintuples $(o_j, f_{jk}, so_{ijkl}, h_i, t_l)$ in a document

Structure the unstructured

- Use in data visualisation tools
- **Quantitative** and **qualitative** analysis

- To answer questions like:
  - Do people like the new iPhone?
  - Among them, what are the features that they most dislike?
  - Among people that don't like the iPhone, what feature is missing?

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- (iPhone, phone, positive, I, 25/10/2018)
- (iPhone, touch screen, positive, I, 25/10/2018)
- (iPhone, voice quality, positive, I, 25/10/2018)
- (iPhone, battery life, negative, I, 25/10/2018)
- (iPhone, price, negative, mother, 25/10/2018)

Sentiment Analysis: not just ONE problem

Discover all quintuples $(o_j, f_{jk}, so_{ijkl}, h_i, t_l)$ in a document

- o_j – the target **object** or **entity**:
  - **Named Entity Recognition**
  - **Coreference resolution**
- f_{jk} – a **feature** or **aspect**: **Information Extraction** (more about that later...)
- so_{ijkl} – **sentiment value**: **Sentiment Identification**
- h_i – **sentiment holder**: **Information/Data Extraction**
- t_l – **time**: **Information/Data Extraction**

Sentiment Analysis: not just ONE problem

Discover all quintuples $(o_j, f_{jk}, so_{ijkl}, h_i, t_l)$ in a document

- **Summarising opinions** is often necessary (e.g. Kindle Paperwhite has over 5K reviews)
- Often reviews talk about **more than one feature**
- Quintuple extraction can help solve this issue

SA granularity levels

Sentiment Analysis granularity $\rightarrow$ document level

- Assumption: document focuses on a **single object** from a **single opinion holder**.
- Goal: discover $(\_, \_, so, \_, \_)$
- ignore o_j, f_{jk}, h_i, t_l
- **Reviews** usually satisfy the assumption
  - Positive = 4 or 5 ☆, Neutral = 3 ☆, Negative = 1 or 2 ☆

Sentiment Analysis granularity → sentence level

- **Sentence** level SA is similar to document level
  - Assumption: sentence contains a **single opinion** from a **single opinion holder**
  - Only an **intermediate** step
  - Consists of two steps:
    1. **Subjectivity** classification → **detect** if the sentence expresses an opinion
    2. **Sentiment** classification → identifies the sentence **polarity**

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   - Corpus-based (Naive Bayes)
2. **Sentiment** classification
   - Rule-based
   - Corpus-based

Sentiment Analysis granularity $\rightarrow$ feature level 1/2

Motivation

- Document and sentences may contain **mixed** opinions
- **Targets** of opinions (**features**) are ignored
- SA at this level provides a **general** opinion on the **object**
- Does not mean that **opinion holder** likes/dislikes everything about it.

- **Feature** level aims to provide a more fine-grained analysis
  - **which** component people like/dislike
  - more informative analysis
  - consists of 5 steps

Sentiment Analysis granularity → feature level 2/2

1. Identify **entities** or **objects**
   - Given a set Q of **seed entities** of class C, and a set D of **candidate entities**, determine which of the entities in D belong to C.
   - binary classification problem

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5. [optional] Produce a summary of all feature-based opinions

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- Explain the differences between subjective and factual data
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- Define two granularity levels for SA

Reading material

- Bing Liu and Lei Zhang (2012). A survey on opinion mining and sentiment analysis. Kluwer Academic Publishers: http://www.cs.uic.edu/~lzhang3/paper/opinion_survey.pdf
- Bing Liu (2010). Sentiment Analysis and Subjectivity. Handbook of Natural Language Processing, Second Edition, (editors: N. Indurkhya and F. J. Damerau): https://www.cs.uic.edu/~liub/FBS/NLP-handbook-sentiment-analysis.pdf

Credits: slides based on Loïc Barrault's slides for the autumn 2020/2021 COM3110/COM4115 course and Bing Liu's slides for AAAI 2011/EACL 2012 tutorials.

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